$$\mathbf{d} : \mathbf{P}^{\mathrm{op}} \times \mathbf{R} \to \mathbf{Bool} \qquad \mathbf{e} : \mathbf{Q}^{\mathrm{op}} \times \mathbf{S} \to \mathbf{Bool},$$

$$\mathbf{d} \otimes \mathbf{e} : (\mathbf{P} \otimes \mathbf{Q})^{\mathrm{op}} \times (\mathbf{R} \otimes \mathbf{S}) \to \mathbf{Bool}$$

$$\langle a^* \mathbin{⨾} c^*,\ b \mathbin{⨾} d \rangle \mapsto \mathbf{d}(a^*, c) \wedge \mathbf{e}(c^*, d)$$